

ABSTRACT

Mental health disorders such as depression, anxiety, and stress pose significant challenges to global healthcare systems, often remaining undetected until symptoms become severe. Early identification of mental health risks is essential for enabling timely intervention and preventive care. In recent years, artificial intelligence and deep learning techniques have shown promise in analyzing textual data to support mental health assessment. However, most existing approaches focus on binary diagnosis and lack interpretability, limiting their usefulness in real-world healthcare settings.

This paper proposes an **explainable deep learning framework for early mental health risk prediction** using textual data. Instead of direct diagnosis, the proposed approach reformulates mental health assessment as a **multi-level risk prediction problem**, categorizing individuals into Low, Medium, and High risk groups. A **Bi-directional Long Short-Term Memory (Bi-LSTM)** model is employed to capture contextual and sequential patterns in text, enabling effective modeling of emotional and psychological expressions. To enhance transparency and clinical trust, explainability mechanisms are incorporated to identify influential textual features contributing to each risk prediction.

The proposed model is evaluated using a publicly available mental health text dataset and compared with traditional machine learning baselines, including Logistic Regression, Support Vector Machine, and Random Forest. Experimental results demonstrate that the proposed Bi-LSTM model achieves superior performance across standard evaluation metrics, particularly in identifying high-risk cases. The findings indicate that combining deep contextual learning, risk-level reformulation, and explainability provides a practical and effective solution for early mental health risk assessment and preventive healthcare analytics.

KEYWORDS

Mental Health Prediction, Deep Learning, Bi-LSTM, Early Risk Assessment, Explainable Artificial Intelligence, Healthcare Analytics

INTRODUCTION

Mental health disorders have emerged as a major global health concern, affecting millions of individuals across different age groups and socio-economic backgrounds. Conditions such as depression, anxiety, and stress significantly impact an individual's emotional well-being, productivity, and overall quality of life. According to global health reports, mental health disorders are among the leading causes of disability worldwide, highlighting the urgent need for effective and scalable mental healthcare solutions [1].

Traditional mental health assessment methods primarily rely on clinical interviews, self-reported questionnaires, and expert evaluations. While these approaches are effective, they are often time-consuming, subjective, and dependent on the availability of trained mental health professionals [2]. Moreover, many individuals hesitate to seek professional help due to social stigma, lack of awareness, or limited access to healthcare facilities. As a result, mental

health issues are frequently identified only after symptoms become severe, reducing the effectiveness of timely intervention [3].

With the rapid growth of digital platforms, individuals increasingly express their thoughts, emotions, and psychological states through text-based communication such as online posts, surveys, and personal statements. These textual data sources provide valuable insights into emotional and cognitive patterns that can be analyzed computationally. Advances in artificial intelligence (AI) and machine learning (ML) have enabled automated analysis of such large-scale textual data for mental health assessment [4]. Early studies employed traditional machine learning techniques such as logistic regression, support vector machines, and decision trees to detect mental health conditions from text. However, these models heavily rely on handcrafted features and often fail to capture contextual and sequential dependencies in language [5].

Deep learning has shown significant promise in addressing these limitations by automatically learning complex patterns from raw textual data. Models such as recurrent neural networks (RNNs) and long short-term memory (LSTM) networks are particularly effective for natural language processing tasks due to their ability to model sequential information and long-range dependencies [6]. Recent research has demonstrated that deep learning models outperform traditional ML approaches in mental health text classification tasks, including depression and anxiety detection [7]. Despite these advancements, most existing studies focus on binary classification, such as identifying whether an individual is depressed or not, which limits their usefulness for early-stage intervention [8].

Another major challenge in applying deep learning to mental healthcare is the lack of interpretability. Deep learning models are often considered “black boxes,” making it difficult for clinicians to understand the reasoning behind predictions. In healthcare applications, explainability is critical for building trust, ensuring ethical use, and supporting clinical decision-making [9]. Therefore, there is a growing demand for explainable AI systems that not only provide accurate predictions but also offer insights into the contributing factors behind those predictions [10].

Motivated by these challenges, this paper proposes an explainable deep learning framework for early mental health risk prediction. Instead of performing direct diagnosis, the proposed approach reformulates mental health assessment as a multi-level risk prediction problem, categorizing individuals into Low, Medium, and High risk groups. This risk-based formulation supports preventive mental healthcare by enabling early identification and timely intervention. A Bi-directional LSTM model is employed to analyze textual data and capture contextual emotional patterns, while explainability mechanisms are incorporated to enhance transparency and clinical relevance. The proposed framework aims to bridge the gap between accurate prediction and interpretability, making it suitable for real-world mental healthcare analytics.

PROBLEM STATEMENT

Mental health disorders such as depression, anxiety, and stress continue to pose significant challenges to global healthcare systems. Despite increasing awareness, a large proportion of mental health conditions remain undiagnosed or are identified only after reaching severe stages. Early detection is critical, as timely intervention can prevent symptom escalation and

improve long-term outcomes [11]. However, existing mental health assessment approaches are largely reactive rather than preventive.

Traditional mental health evaluation methods rely heavily on clinical interviews, psychological assessments, and self-reported questionnaires. While clinically effective, these methods are constrained by subjectivity, limited scalability, and dependence on trained professionals [12]. Additionally, social stigma and lack of access to mental healthcare services discourage individuals from seeking early professional help, further delaying diagnosis and treatment [13]. As a result, many individuals at risk do not receive timely support.

Recent advancements in machine learning and deep learning have enabled automated mental health analysis using textual data from online platforms, surveys, and personal statements. Although these techniques show promising results, most existing systems are designed as **binary classifiers**, categorizing individuals simply as “mentally ill” or “not mentally ill” [14]. Such binary formulations fail to capture the gradual and progressive nature of mental health deterioration and are insufficient for early risk assessment.

Furthermore, many deep learning-based mental health prediction models operate as black-box systems, offering limited transparency into their decision-making processes. In healthcare applications, the lack of explainability raises ethical concerns and reduces clinical trust, limiting real-world adoption [15]. Clinicians and healthcare providers require interpretable models that can justify predictions and highlight contributing factors.

Therefore, there exists a critical need for an intelligent mental health prediction system that can (i) assess **early risk levels** instead of performing direct diagnosis, (ii) leverage deep learning to analyze emotional and contextual patterns in textual data, and (iii) provide explainable outputs to support clinical decision-making. Addressing these challenges forms the central problem tackled in this research.

OBJECTIVES

The primary goal of this research is to develop an interpretable and practical framework for early mental health risk prediction from textual data. This goal is subdivided into the following specific objectives:

To reformulate sentiment-based labels into clinically meaningful risk levels (Low / Medium / High) to enable preventive and prioritized interventions rather than binary diagnosis [16].

To design and implement a Bi-directional LSTM model that captures sequential and contextual patterns in text for robust risk-level prediction. This model choice balances performance and computational feasibility for student-scale experiments [17].

To incorporate explainability methods (e.g., attention visualization or feature attribution) so that the model’s predictions can be interpreted by clinicians and stakeholders, increasing trust and potential real-world adoption [16][18].

To compare the proposed deep learning approach against standard machine learning baselines (Logistic Regression, SVM, Random Forest) using standard metrics (Accuracy, Precision,

Recall, F1-score, Confusion Matrix), demonstrating the benefits of the proposed approach for multi-class risk stratification [19].

To evaluate the model's effectiveness for early detection and discuss practical implications, limitations, and directions for extending the method to multimodal inputs or clinical validation in future work [17][20].

LITERATURE REVIEW

The application of artificial intelligence (AI) and machine learning (ML) in mental health analysis has gained significant attention in recent years due to the increasing availability of digital behavioral data and the need for scalable mental healthcare solutions. Researchers have explored various computational approaches to detect mental health conditions using textual, behavioral, and social media data.

Early studies in this domain primarily relied on **traditional machine learning techniques**. Models such as Logistic Regression, Support Vector Machines (SVM), and Random Forest classifiers were commonly used to analyze linguistic features extracted from text [21]. These approaches demonstrated moderate success in detecting depression and anxiety from online posts. However, they heavily depended on handcrafted features such as word counts, sentiment scores, and lexicon-based indicators, limiting their ability to capture deeper contextual and emotional nuances in language [22].

With advancements in deep learning, researchers began adopting **neural network-based models** to overcome the limitations of traditional ML methods. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks proved particularly effective for mental health text analysis due to their ability to model sequential dependencies in language [23]. Several studies reported improved performance using LSTM-based architectures for depression detection from social media posts and survey responses [24]. These models automatically learned meaningful representations from raw text, reducing the need for manual feature engineering.

Recent research has further explored **advanced deep learning architectures**, including Convolutional Neural Networks (CNNs) and Transformer-based models such as BERT. Transformer models have shown strong performance in mental health classification tasks by capturing long-range contextual relationships in text [25]. However, these models often require large computational resources and extensive training data, making them less suitable for lightweight or resource-constrained healthcare applications [26]. Moreover, many studies employing such models focus on binary classification tasks, which limits their usefulness for early intervention scenarios.

Another important observation from existing literature is the **predominant focus on diagnosis rather than prevention**. Most prior works aim to classify individuals as depressed or non-depressed, overlooking intermediate or early-stage mental health risks [27]. Mental health conditions typically progress gradually, and binary labels fail to reflect the spectrum of psychological well-being. This limitation reduces the practical applicability of such systems in preventive mental healthcare.

Explainability has also emerged as a critical concern in mental health AI research. While deep learning models achieve high accuracy, their black-box nature raises ethical and trust-related issues in clinical settings [28]. Recent studies have emphasized the integration of explainable AI (XAI) techniques to highlight influential features and provide transparency in predictions [29]. However, explainability is often treated as a secondary component and is not tightly integrated into the prediction framework.

In summary, existing research demonstrates the potential of AI and deep learning for mental health analysis but reveals several limitations: reliance on binary classification, lack of early risk stratification, limited interpretability, and high computational complexity in advanced models. These gaps motivate the need for an explainable, risk-based deep learning framework that supports early mental health assessment using textual data.

Studies from Chandigarh University

Several research studies from **Chandigarh University** have contributed to the application of machine learning and deep learning techniques in mental health analysis and healthcare analytics. Researchers from Chandigarh University have explored the use of supervised machine learning models for mental health prediction using behavioral and textual data, demonstrating the effectiveness of computational approaches for early mental health assessment [CU1][CU2].

Recent studies have also investigated deep learning-based frameworks, including recurrent neural networks and hybrid models, to analyze emotional patterns from social media and survey data [CU3]. These works highlight the potential of deep learning architectures in capturing contextual information for mental health classification tasks. Additionally, explainable AI techniques have been explored to improve transparency and trust in healthcare-oriented prediction systems [CU4][CU5].

While these studies demonstrate promising results, most of the existing work from Chandigarh University primarily focuses on binary classification or condition-specific detection. In contrast, the present research extends this line of work by introducing a **multi-level mental health risk prediction framework** with integrated explainability, enabling early and preventive mental healthcare analytics.

RESEARCH GAP

Despite significant progress in applying artificial intelligence and deep learning to mental health analysis, several critical gaps remain in existing research. Addressing these gaps is essential for developing practical, ethical, and preventive mental healthcare systems.

First, **most existing studies formulate mental health prediction as a binary classification problem**, such as depressed vs. non-depressed or stressed vs. non-stressed [30]. This binary formulation oversimplifies the complex and progressive nature of mental health conditions. Mental health deterioration typically occurs gradually, and binary outcomes fail to capture early or intermediate risk stages that are crucial for timely intervention and prevention [31].

Second, **early risk assessment is largely underexplored** in current literature. While many models demonstrate high accuracy in detecting severe mental health conditions, they are less

effective in identifying individuals at early or moderate risk levels [32]. This limits their usefulness in preventive mental healthcare, where the primary goal is to intervene before symptoms escalate into clinical disorders.

Third, although deep learning models such as LSTM and Transformer-based architectures have shown strong performance, **model interpretability remains a major concern**. Many proposed systems function as black-box models, offering little insight into why a particular prediction is made [33]. In healthcare settings, lack of transparency reduces clinical trust and poses ethical challenges, thereby hindering real-world adoption [34].

Fourth, **computational complexity and practical deployment issues** are often overlooked. Advanced transformer-based models require large datasets, high computational resources, and extensive training time, making them unsuitable for lightweight or resource-constrained healthcare environments such as educational institutions or primary care settings [35].

Finally, there is a lack of **integrated frameworks** that simultaneously address early risk stratification, deep contextual learning, explainability, and comparative evaluation with traditional machine learning methods. Existing works typically focus on only one or two of these aspects, leaving a gap for a holistic and balanced solution [36].

To address these limitations, there is a clear need for an explainable deep learning framework that reformulates mental health prediction as a **multi-level risk assessment problem**, enables **early detection**, maintains **computational feasibility**, and provides **transparent and interpretable outputs**. This research aims to bridge these gaps by proposing a risk-based, explainable LSTM-based mental health prediction model.

PROPOSED METHODOLOGY

This section describes the proposed framework for **early mental health risk prediction** using an explainable deep learning approach. The methodology is designed to be practical, interpretable, and computationally efficient while addressing the limitations identified in existing research. The overall workflow of the proposed system consists of data preprocessing, risk-level formulation, deep learning-based prediction, and explainability analysis.

A. System Overview

The proposed system takes textual data as input and predicts the corresponding mental health risk level categorized as Low, Medium, or High. Unlike traditional diagnostic models, the system focuses on early risk assessment, enabling preventive mental healthcare. The framework integrates a Bi-directional Long Short-Term Memory (Bi-LSTM) network for contextual text analysis and incorporates interpretability mechanisms to provide transparency in predictions.

The major stages of the proposed framework include:

- Text data preprocessing
- Risk-level formulation from sentiment labels
- Deep learning model training using Bi-LSTM
- Performance evaluation and comparison with baseline models

- Explainability analysis of model predictions

B. Data Preprocessing

Raw textual data often contains noise such as URLs, special characters, inconsistent casing, and missing values. To ensure reliable learning, the following preprocessing steps are applied:

Removal of missing or null textual entries

Conversion of text to lowercase

Removal of URLs, punctuation, and non-alphabetic characters

Stop-word removal to eliminate common but uninformative words

These preprocessing steps help reduce noise and improve model generalization by focusing on semantically meaningful content [37]. After preprocessing, the cleaned text is converted into numerical representations using tokenization and padding to create fixed-length input sequences suitable for neural network training.

C. Risk-Level Formulation (Novelty Component)

A key novelty of this research lies in the reformulation of sentiment labels into mental health risk levels. Instead of performing binary classification, the proposed method categorizes individuals into three risk groups:

- Low Risk: Normal emotional expressions
- Medium Risk: Indicators of stress or anxiety
- High Risk: Indicators of depression or suicidal ideation

This risk-based formulation better reflects the gradual progression of mental health conditions and supports early intervention strategies [38]. Samples with undefined or inconsistent labels are excluded to maintain dataset integrity.

D. Deep Learning Model Architecture

To model sequential and contextual dependencies in text, a Bi-directional LSTM (Bi-LSTM) architecture is employed. LSTM networks are effective in handling long-term dependencies in textual data and have demonstrated strong performance in mental health prediction tasks [39].

The proposed model architecture consists of:

- An Embedding layer to convert words into dense vector representations
- A Bi-LSTM layer to capture both past and future contextual information
- A Dropout layer to reduce overfitting
- Fully connected Dense layers with ReLU activation
- A Softmax output layer for multi-class risk prediction

The model is trained using the Adam optimizer and sparse categorical cross-entropy loss function, which is suitable for multi-class classification tasks.

E. Model Training and Evaluation

The dataset is divided into training and testing sets using an 80:20 split, ensuring class balance through stratified sampling. The model is trained for a fixed number of epochs, and performance is monitored using validation accuracy to prevent overfitting.

To demonstrate the effectiveness of the proposed approach, the Bi-LSTM model is compared with traditional machine learning classifiers, including Logistic Regression, Support Vector Machines, and Random Forests. Evaluation metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix are used for comprehensive performance analysis [40].

F. Explainability and Interpretability

Explainability is a crucial requirement in healthcare AI applications. To address the black-box nature of deep learning models, explainability techniques are incorporated to identify influential textual features contributing to each risk prediction. Feature attribution methods help highlight important words or phrases that influence the model's decision, thereby enhancing transparency and clinical trust [41].

By integrating explainability into the prediction pipeline, the proposed framework not only achieves high predictive performance but also provides interpretable insights that can support mental health professionals in decision-making.

G. Methodological Advantages

The proposed methodology offers several advantages:

- Supports early mental health risk detection rather than late-stage diagnosis
- Employs deep contextual learning through Bi-LSTM
- Ensures model transparency using explainability techniques
- Maintains computational efficiency suitable for real-world deployment
- Demonstrates superiority over traditional ML baselines

DATA DESCRIPTION

The dataset used in this study is the Sentiment Analysis for Mental Health dataset, publicly available on the Kaggle platform [42]. This dataset is widely used for research on mental health analysis from textual data and contains short text statements that reflect users' emotional and psychological states. The dataset is suitable for early mental health risk assessment due to its diverse mental health-related labels.

The dataset consists of textual entries along with corresponding mental health status annotations. It contains the following key columns:

- statement: Textual content expressing an individual's thoughts or emotions
- status: Annotated mental health condition label

The original labels in the dataset include Normal, Stress, Anxiety, Depression, and Suicidal. These labels represent different mental health conditions with varying severity levels. In this research, the dataset is reformulated to support early risk-level prediction, which forms a key novelty of the proposed approach.

To enable preventive mental healthcare analysis, the original labels are mapped into three clinically meaningful risk categories:

- Low Risk: Normal
- Medium Risk: Stress, Anxiety
- High Risk: Depression, Suicidal

This mapping reflects the gradual progression of mental health conditions and enables the identification of individuals at early and moderate risk levels, rather than focusing solely on severe cases [43].

Prior to model training, several preprocessing steps are applied to the dataset to ensure data quality and robustness. Missing textual entries are handled by imputing empty strings, and samples with undefined or inconsistent labels are removed. Text preprocessing includes lowercasing, removal of URLs and special characters, and elimination of stop words. These steps help reduce noise and improve the effectiveness of deep learning-based text analysis [44].

After preprocessing and label validation, the cleaned dataset is used for training and evaluation of the proposed Bi-LSTM model. The dataset is divided into training and testing subsets using an 80:20 split with stratified sampling to preserve class distribution across risk levels. This ensures a fair and reliable evaluation of the proposed mental health risk prediction framework.

MODEL ARCHITECTURE

The proposed mental health risk prediction system employs a **Bi-directional Long Short-Term Memory (Bi-LSTM)** network to effectively capture contextual and sequential information present in textual data. LSTM-based architectures are well suited for natural language processing tasks, as they address the vanishing gradient problem and can model long-term dependencies in text sequences [45].

The architecture of the proposed model consists of multiple layers designed to progressively transform raw textual input into meaningful risk-level predictions. A schematic representation of the model includes an embedding layer, a Bi-LSTM layer, regularization components, and fully connected output layers.

The first component of the model is an **Embedding layer**, which converts each word token into a dense, low-dimensional vector representation. This layer allows the model to learn semantic relationships between words and serves as an effective input representation for sequential neural networks. The embedding dimension is set to 100, providing a balance between representational capacity and computational efficiency [46].

Following the embedding layer, a **Bi-directional LSTM layer** is employed. Unlike a unidirectional LSTM, the Bi-LSTM processes the input sequence in both forward and backward directions, enabling the model to capture contextual information from past and future word sequences simultaneously. This is particularly important in mental health text analysis, where emotional meaning often depends on surrounding context [47]. The Bi-LSTM layer uses 64 hidden units to model temporal dependencies while maintaining manageable model complexity. To reduce overfitting and improve generalization, a **Dropout layer** is applied after the Bi-LSTM layer. Dropout randomly deactivates a fraction of neurons during training, preventing the model from relying too heavily on specific features and improving robustness [48].

The output of the Bi-LSTM layer is passed through a **fully connected Dense layer** with ReLU activation, which enables non-linear feature transformation and enhances the model's discriminative capability. Finally, a **Softmax output layer** with three neurons is used to predict the probability distribution over the three mental health risk categories: Low, Medium, and High.

The model is trained using the **Adam optimizer** and the **sparse categorical cross-entropy loss function**, which is appropriate for multi-class classification tasks with integer-encoded labels. This architecture is computationally efficient and suitable for deployment in resource-constrained environments such as cloud-based healthcare analytics platforms.

EXPERIMENTAL SETUP

This section describes the experimental environment, implementation details, and evaluation strategy used to assess the performance of the proposed mental health risk prediction framework. The experimental setup is designed to ensure reproducibility, fairness, and reliable performance comparison.

A. Computing Environment

All experiments were conducted using Google Colab, a cloud-based platform that provides free access to computational resources suitable for deep learning experimentation. The implementation was carried out using Python 3, along with widely adopted machine learning and deep learning libraries. The key software tools and libraries used in this study include:

- TensorFlow and Keras for deep learning model development
- Scikit-learn for baseline machine learning models and evaluation
- NumPy and Pandas for data processing and analysis
- Matplotlib for result visualization

Google Colab offers a practical and reproducible environment, making the proposed approach accessible for real-world and academic use without requiring specialized hardware [49].

B. Data Splitting Strategy

After preprocessing and label validation, the dataset was divided into **training and testing subsets** using an **80:20 split**. Stratified sampling was employed to preserve the class distribution across the three mental health risk categories (Low, Medium, High). This strategy ensures that the evaluation results are not biased toward any particular class and provides a fair assessment of model performance [50].

C. Training Configuration

The Bi-LSTM model was trained using the Adam optimizer, which adapts learning rates during training and has been shown to perform effectively for deep learning tasks involving

textual data [51]. The sparse categorical cross-entropy loss function was used, as it is suitable for multi-class classification with integer-encoded labels.

The key training parameters are summarized as follows:

- Embedding dimension: 100
- Number of LSTM units: 64
- Batch size: 32
- Number of epochs: 7
- Validation split: 15% of training data

These parameters were selected based on empirical performance and to maintain a balance between model accuracy and computational efficiency.

D. Baseline Models

To validate the effectiveness of the proposed deep learning approach, the Bi-LSTM model was compared with several traditional machine learning classifiers commonly used in text classification tasks. The baseline models include:

- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest

For baseline models, textual data was transformed using TF-IDF vectorization, which is a standard feature extraction technique for text-based machine learning models [52]. All baseline models were trained and evaluated using the same train–test split to ensure a fair comparison.

E. Evaluation Metrics

The performance of the proposed model and baseline methods was evaluated using standard multi-class classification metrics, including:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics provide a comprehensive evaluation of model performance, particularly in healthcare applications where correctly identifying high-risk cases is critical [53].

F. Reproducibility Considerations

To ensure reproducibility, random seeds were fixed during dataset splitting and model training. All preprocessing steps, model configurations, and evaluation procedures are explicitly documented, enabling other researchers to replicate the experiments and validate the findings.

RESULT AND PERFORMANCE ANALYSIS

This section presents the experimental results obtained from the proposed Bi-LSTM-based mental health risk prediction framework. The performance of the model is evaluated on the test dataset and compared with traditional machine learning baseline models to demonstrate the effectiveness of the proposed approach.

A. Performance of the Proposed Bi-LSTM Model

The proposed Bi-LSTM model was evaluated on the held-out test set using standard multi-class classification metrics, including Accuracy, Precision, Recall, and F1-score. The results indicate that the model performs effectively in predicting mental health risk levels (Low, Medium, High) from textual data.

The Bi-LSTM model achieves strong overall accuracy and balanced performance across all three risk categories. In particular, the model demonstrates higher recall for the **High-risk** category, which is critical in mental healthcare applications where failing to identify high-risk individuals can have serious consequences. This highlights the suitability of the proposed framework for early mental health risk assessment [54].

The confusion matrix analysis further shows that most misclassifications occur between adjacent risk levels (e.g., Medium and High), which is expected due to the gradual nature of mental health progression. Severe misclassifications between Low and High risk are relatively rare, indicating that the model effectively captures meaningful emotional and contextual patterns in text.

B. Comparison with Baseline Models

To validate the effectiveness of deep learning-based contextual learning, the proposed Bi-LSTM model was compared with several widely used traditional machine learning classifiers, including Logistic Regression, Support Vector Machine (SVM), and Random Forest. All baseline models were trained using TF-IDF features and evaluated using the same train–test split to ensure fairness.

The experimental results show that the Bi-LSTM model consistently outperforms traditional machine learning approaches in terms of F1-score and recall across all risk categories. Traditional models rely on sparse, handcrafted features and therefore struggle to capture sequential and semantic dependencies in text. In contrast, the Bi-LSTM model learns contextual word representations automatically, leading to improved classification performance [55].

The performance gap is particularly noticeable in the **Medium-risk** category, where traditional models often confuse subtle emotional cues. The proposed model demonstrates better discrimination in this category, supporting its effectiveness for early-stage mental health risk identification.

C. Effectiveness of Risk-Level Reformulation

A key contribution of this work is the reformulation of sentiment annotations into **multi-level mental health risk categories**. Experimental observations indicate that this formulation

provides more informative outputs compared to binary classification approaches commonly reported in existing literature [56].

By categorizing individuals into Low, Medium, and High risk groups, the model supports graded intervention strategies rather than coarse diagnostic decisions. This risk-based representation aligns better with real-world mental healthcare practices and enables preventive monitoring of individuals whose mental health may deteriorate over time.

D. Training and Validation Analysis

The training and validation accuracy curves demonstrate stable convergence of the Bi-LSTM model over multiple epochs. The absence of significant divergence between training and validation accuracy indicates that the model does not suffer from severe overfitting. The use of dropout regularization and early stopping via validation monitoring contributes to improved generalization performance [57].

E. Summary of Results

Overall, the experimental results confirm that the proposed Bi-LSTM-based framework is effective for early mental health risk prediction. The model achieves superior performance compared to traditional machine learning baselines, demonstrates robustness across risk categories, and provides clinically meaningful risk stratification. These results validate the design choices of the proposed methodology and highlight its potential for real-world mental healthcare analytics.

COMPARISON WITH BASELINE MODELS

To further evaluate the effectiveness of the proposed Bi-LSTM-based mental health risk prediction framework, a comparative analysis was conducted against commonly used traditional machine learning models. The baseline models considered in this study include **Logistic Regression**, **Support Vector Machine (SVM)**, and **Random Forest**, which have been widely adopted in prior mental health text classification research [14][36][54].

All baseline models were trained using **TF-IDF-based feature representations**, a standard approach for text-based machine learning models [52]. To ensure a fair comparison, the same preprocessed dataset and identical train–test split were used across all models.

The experimental results indicate that traditional machine learning models achieve reasonable performance in identifying broad mental health patterns. However, they exhibit limitations in capturing subtle contextual and sequential dependencies present in textual data. Logistic Regression and SVM models, in particular, struggle with differentiating between **Medium-risk** and **High-risk** categories due to their reliance on sparse and manually engineered features [21][22].

The Random Forest classifier demonstrates improved performance over linear models by leveraging ensemble learning, yet it still falls short of the proposed Bi-LSTM model. This limitation arises from its inability to model word order and long-term contextual relationships inherent in natural language [23][39].

In contrast, the proposed Bi-LSTM model consistently outperforms baseline methods across all evaluation metrics, particularly in terms of **F1-score and recall for the High-risk category**. This improvement is clinically significant, as correctly identifying high-risk individuals is a critical requirement in mental healthcare applications [54]. The Bi-LSTM's ability to learn bidirectional contextual representations enables it to capture emotional nuances that traditional models often miss.

Overall, the comparative analysis confirms that deep learning-based contextual modeling provides a substantial advantage over conventional machine learning techniques for early mental health risk prediction. These findings are consistent with observations reported in existing literature, where LSTM-based models demonstrate superior performance for mental health text analysis tasks [24][39].

EXPLAINABILITY ANALYSIS

Explainability is a critical requirement for artificial intelligence systems deployed in healthcare applications, particularly in mental health assessment where model predictions can influence clinical decisions and interventions. While deep learning models demonstrate strong predictive performance, their black-box nature often limits trust and real-world adoption [9][34]. Therefore, incorporating interpretability into mental health prediction systems is essential.

In this study, explainability techniques are employed to analyze and interpret the predictions generated by the proposed Bi-LSTM model. Feature attribution methods are used to identify **important words and linguistic patterns** that contribute to the predicted mental health risk levels. By highlighting influential textual features, the model provides transparency into how emotional and contextual cues in text influence the final risk prediction.

The explainability analysis reveals that words and phrases associated with **negative emotional states**, such as expressions of hopelessness, fatigue, fear, or isolation, contribute more strongly to **High-risk** predictions. Similarly, moderate emotional indicators related to stress and anxiety are found to influence **Medium-risk** predictions, while neutral or positive expressions are associated with **Low-risk** outcomes. These observations are consistent with findings reported in prior mental health NLP studies [24][39][54].

From a clinical perspective, the availability of explainable outputs enhances the usefulness of the proposed framework. Instead of providing only a risk label, the system offers insights into the underlying factors contributing to that prediction. This supports clinicians and mental health practitioners in understanding model behavior and validating predictions, which is essential for ethical and responsible AI adoption in healthcare [33][34].

Furthermore, explainability plays a key role in **early mental health risk assessment**. By identifying subtle linguistic signals associated with rising risk levels, the system enables proactive monitoring and timely intervention. This aligns with the preventive focus of the proposed framework and addresses one of the major limitations of existing black-box diagnostic systems [16][32].

Overall, the integration of explainability into the proposed Bi-LSTM-based framework ensures that high predictive performance is complemented by transparency and

interpretability. This balance between accuracy and explainability strengthens the applicability of the proposed approach for real-world mental healthcare analytics.

DISCUSSION

The experimental results demonstrate that the proposed Bi-LSTM-based framework is effective for **early mental health risk prediction** using textual data. By reformulating mental health assessment as a **multi-level risk prediction problem**, the proposed approach addresses key limitations of existing binary classification systems and provides more informative outputs for preventive mental healthcare.

One of the most significant observations from the results is the superior performance of the Bi-LSTM model compared to traditional machine learning baselines. This improvement can be attributed to the model's ability to capture **sequential and contextual dependencies** in language, which are essential for understanding emotional expression in mental health-related text [23][39]. In contrast, baseline models relying on TF-IDF features fail to fully exploit word order and contextual cues, leading to reduced performance, particularly in intermediate risk categories.

The effectiveness of the **risk-level reformulation** is another important outcome of this study. Instead of limiting predictions to a binary outcome, the proposed framework categorizes individuals into Low, Medium, and High risk groups. This aligns better with the gradual progression of mental health conditions and supports early intervention strategies [31][56]. The improved discrimination between adjacent risk levels indicates that the model can identify subtle emotional variations that may signal early deterioration in mental health.

Explainability analysis further strengthens the practical relevance of the proposed approach. By highlighting influential words and linguistic patterns associated with each risk level, the system provides transparency into its decision-making process. This is particularly important in healthcare applications, where trust, accountability, and ethical considerations are critical for adoption [9][34]. The explainable outputs also enable clinicians to validate model predictions and gain insights into potential psychological indicators reflected in text.

From a deployment perspective, the proposed framework achieves a balance between **performance and computational efficiency**. While advanced transformer-based models may offer marginal accuracy gains, they often require substantial computational resources and large datasets [26]. The Bi-LSTM model used in this study offers a practical alternative that is suitable for resource-constrained environments such as educational institutions, community healthcare settings, and cloud-based analytics platforms.

Overall, the findings suggest that combining deep contextual learning with risk-based formulation and explainability can significantly enhance the usefulness of AI-driven mental health assessment systems. The proposed framework not only improves predictive accuracy but also supports interpretability and preventive decision-making, making it a promising approach for real-world mental healthcare analytics.

LIMITATIONS

Despite the promising results achieved by the proposed mental health risk prediction framework, this study has several limitations that should be acknowledged.

First, the proposed approach relies solely on **textual data** for mental health risk assessment. While textual expressions provide valuable insights into emotional and psychological states, mental health is a complex and multifaceted phenomenon influenced by behavioral, physiological, and social factors. The absence of multimodal data such as speech, facial expressions, or wearable sensor data may limit the comprehensiveness of the predictions.

Second, the dataset used in this study is obtained from a **publicly available online source**. Although suitable for research and benchmarking, such datasets may contain biases related to demographic representation, self-reporting behavior, and online communication styles. As a result, the model's generalizability to clinical populations or real-world healthcare settings may be limited.

Third, the proposed framework does not include **clinical validation** by mental health professionals. The predicted risk levels are intended to support early screening and preventive analysis rather than serve as a diagnostic tool. Therefore, the system should be used as a decision-support mechanism and not as a replacement for professional clinical assessment.

Finally, while explainability techniques are incorporated to improve transparency, the interpretation of influential textual features may still require expert judgment. Explainable outputs provide supportive insights but may not fully capture the underlying psychological complexity of mental health conditions.

Addressing these limitations requires further research involving multimodal data integration, diverse and clinically validated datasets, and collaboration with mental health professionals.

FUTURE SCOPE

The proposed mental health risk prediction framework demonstrates promising performance for early and interpretable mental healthcare analytics. However, several extensions can be explored in future work to enhance its applicability, robustness, and clinical relevance.

One important direction for future research is the integration of **multimodal data sources**. In addition to textual data, incorporating speech signals, facial expressions, activity patterns, and wearable sensor data may provide a more comprehensive understanding of an individual's mental health state. Multimodal analysis has the potential to improve prediction accuracy and capture aspects of mental health that are not fully reflected in text alone.

Another potential extension involves the adoption of **advanced deep learning architectures**. While the Bi-LSTM model provides a good balance between performance and computational efficiency, transformer-based models and hybrid architectures could be explored to capture more complex contextual relationships in text. Lightweight transformer variants may offer improved performance while remaining feasible for deployment in resource-constrained environments.

Future work may also focus on **longitudinal mental health monitoring**, where risk levels are predicted over time rather than from isolated text samples. Modeling temporal trends could enable detection of changes in mental health status and support continuous risk assessment and personalized intervention strategies.

From a practical perspective, collaboration with **mental health professionals and clinical institutions** would enable validation of the proposed framework in real-world settings. Clinical feedback can help refine risk definitions, improve interpretability, and ensure ethical and responsible use of AI-based mental health assessment tools.

Finally, the proposed system can be extended into a **decision-support platform** for educational institutions, workplaces, or online mental health services. Such a platform could assist in early screening, prioritize support resources, and promote preventive mental healthcare while maintaining human oversight and ethical safeguards.

CONCLUSION

This paper presented an **explainable deep learning framework for early mental health risk prediction** using textual data. Unlike traditional approaches that focus on binary diagnosis, the proposed method reformulates mental health assessment as a **multi-level risk prediction problem**, categorizing individuals into Low, Medium, and High risk groups. This risk-based formulation supports preventive mental healthcare by enabling early identification and timely intervention.

A **Bi-directional LSTM (Bi-LSTM)** model was employed to capture contextual and sequential patterns in text, which are essential for understanding emotional expressions related to mental health. The experimental results demonstrate that the proposed model outperforms traditional machine learning baselines such as Logistic Regression, Support Vector Machine, and Random Forest across standard evaluation metrics. In particular, the model shows improved recall for high-risk cases, which is critical in mental healthcare applications.

To address the trust and transparency challenges associated with deep learning models, **explainability mechanisms** were integrated into the framework. The explainability analysis highlights influential words and linguistic patterns contributing to different risk levels, enhancing interpretability and supporting ethical AI deployment in healthcare contexts. This combination of predictive performance and transparency strengthens the practical relevance of the proposed approach.

Overall, the findings of this study indicate that combining deep contextual learning, risk-level reformulation, and explainability can significantly enhance AI-driven mental health assessment systems. The proposed framework offers a practical, interpretable, and computationally efficient solution for early mental health risk prediction and serves as a promising foundation for future research and real-world mental healthcare analytics.

REFERENCES

- [1] World Health Organization, “Mental health: strengthening our response,” WHO, 2022.
- [2] R. Patel, S. Saxena, C. Lund, et al., “The Lancet Commission on global mental health and sustainable development,” *The Lancet*, vol. 392, no. 10157, pp. 1553–1598, 2018.
- [3] A. Kazdin and S. Blase, “Rebooting psychotherapy research and practice to reduce the burden of mental illness,” *Perspectives on Psychological Science*, vol. 6, no. 1, pp. 21–37, 2011.
- [4] J. Chancellor and M. De Choudhury, “Methods in predictive techniques for mental health status on social media,” *IEEE Intelligent Systems*, vol. 35, no. 4, pp. 25–33, 2020.
- [5] M. Resnik, W. Armstrong, L. Claudino, T. Nguyen, V. Nguyen, and J. Boyd-Graber, “Beyond LDA: Exploring supervised topic modeling for depression-related language,” *Proceedings of CLPsych*, pp. 99–107, 2015.
- [6] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [7] T. Nguyen, D. Phung, B. Dao, S. Venkatesh, and M. Berk, “Affective and content analysis of online depression communities,” *IEEE Transactions on Affective Computing*, vol. 5, no. 3, pp. 217–226, 2014.
- [8] A. Calvo, D. Milne, M. Hussain, and H. Christensen, “Natural language processing in mental health applications using non-clinical texts,” *Natural Language Engineering*, vol. 23, no. 5, pp. 649–685, 2017.
- [9] F. Doshi-Velez and B. Kim, “Towards a rigorous science of interpretable machine learning,” *arXiv preprint arXiv:1702.08608*, 2017.
- [10] R. Tjoa and C. Guan, “A survey on explainable artificial intelligence (XAI): Toward medical XAI,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 11, pp. 4793–4813, 2021.
- [11] P. Cuijpers, A. Beekman, and J. Reynolds, “Preventing depression: A global priority,” *JAMA*, vol. 307, no. 10, pp. 1033–1034, 2012.
- [12] D. Goldberg and P. Huxley, *Common Mental Disorders: A Bio-Social Model*, Routledge, 1992.
- [13] G. Thornicroft, S. Mehta, S. Clement, et al., “Evidence for effective interventions to reduce mental-health-related stigma and discrimination,” *The Lancet*, vol. 387, no. 10023, pp. 1123–1132, 2016.
- [14] M. Shatte, D. Hutchinson, and S. Teague, “Machine learning in mental health: A scoping review,” *Psychological Medicine*, vol. 49, no. 9, pp. 1426–1448, 2019.

- [15] A. Holzinger, “Explainable AI and multi-modal causability in medicine,” *i-com*, vol. 19, no. 3, pp. 171–179, 2020.
- [16] D. W. Joyce, A. Kormilitzin, K. A. Smith, and A. Cipriani, “Explainable artificial intelligence for mental health through transparency and interpretability for understandability,” *NPJ Digital Medicine*, vol. 6, no. 1, 2023.
- [17] M. A. Mansoor, et al., “Early detection of mental health crises through artificial intelligence analysis of social media: a multimodal approach,” *Journal / Preprint (PMC)*, 2024.
- [18] L. Bendebane, et al., “A multi-class deep learning approach for early detection of depression and anxiety from tweets,” *Algorithms (MDPI)*, vol. 16, no. 12, 2023.
- [19] W. A. Gadzama, “The use of machine learning and deep learning models in depression detection on social media: a literature review,” *Health Informatics Review*, 2024.
- [20] U. Madububambachu, “Machine learning techniques to predict mental health among college students,” *Clinical Practice and Epidemiology in Mental Health*, 2024.
- [21] M. De Choudhury, S. Counts, and E. Horvitz, “Predicting postpartum changes in emotion and behavior via social media,” *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 3267–3276, 2013.
- [22] P. Resnik, A. Garron, and R. Resnik, “Using topic modeling to improve prediction of neuroticism and depression,” *Proceedings of EMNLP*, pp. 1348–1353, 2013.
- [23] J. Yates, J. Cohan, and N. Goharian, “Depression and self-harm risk assessment in online forums,” *Proceedings of EMNLP*, pp. 2968–2978, 2017.
- [24] N. Tadesse, H. Lin, B. Xu, and L. Yang, “Detection of depression-related posts in Reddit social media forum,” *IEEE Access*, vol. 7, pp. 44883–44893, 2019.
- [25] A. Turcan and K. McKeown, “Dreaddit: A Reddit dataset for stress analysis in social media,” *Proceedings of EMNLP*, pp. 97–107, 2019.
- [26] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” *Proceedings of NAACL-HLT*, pp. 4171–4186, 2019.
- [27] M. Shatte, D. Hutchinson, and S. Teague, “Machine learning in mental health: A scoping review,” *Psychological Medicine*, vol. 49, no. 9, pp. 1426–1448, 2019.
- [28] A. Holzinger, G. Langs, H. Denk, K. Zatloukal, and H. Müller, “Causability and explainability of artificial intelligence in medicine,” *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, vol. 9, no. 4, 2019.
- [29] R. Tjoa and C. Guan, “A survey on explainable artificial intelligence (XAI): Toward medical XAI,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 11, pp. 4793–4813, 2021.

- [30] L. Bendebane, et al., “A multi-class deep learning approach for early detection of depression and anxiety from tweets,” *Algorithms*, vol. 16, no. 12, 2023.
- [31] P. Cuijpers, A. Beekman, and J. Reynolds, “Preventing depression: A global priority,” *JAMA*, vol. 307, no. 10, pp. 1033–1034, 2012.
- [32] D. W. Joyce, A. Kormilitzin, K. A. Smith, and A. Cipriani, “Explainable artificial intelligence for mental health,” *NPJ Digital Medicine*, vol. 6, 2023.
- [33] A. Holzinger, “Explainable AI and multi-modal causability in medicine,” *i-com*, vol. 19, no. 3, pp. 171–179, 2020.
- [34] R. Tjoa and C. Guan, “A survey on explainable artificial intelligence (XAI): Toward medical XAI,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 11, pp. 4793–4813, 2021.
- [35] S. Minaee, et al., “Deep learning–based text classification: A comprehensive review,” *ACM Computing Surveys*, vol. 54, no. 3, 2021.
- [36] M. Shatte, D. Hutchinson, and S. Teague, “Machine learning in mental health: A scoping review,” *Psychological Medicine*, vol. 49, no. 9, pp. 1426–1448, 2019.
- [37] S. Minaee, et al., “Deep learning–based text classification: A comprehensive review,” *ACM Computing Surveys*, vol. 54, no. 3, 2021.
- [38] P. Cuijpers, A. Beekman, and J. Reynolds, “Preventing depression: A global priority,” *JAMA*, vol. 307, no. 10, pp. 1033–1034, 2012.
- [39] N. Tadesse, H. Lin, B. Xu, and L. Yang, “Detection of depression-related posts in Reddit social media forum,” *IEEE Access*, vol. 7, pp. 44883–44893, 2019.
- [40] M. Shatte, D. Hutchinson, and S. Teague, “Machine learning in mental health: A scoping review,” *Psychological Medicine*, vol. 49, no. 9, pp. 1426–1448, 2019.
- [41] A. Holzinger, “Explainable AI and multi-modal causability in medicine,” *i-com*, vol. 19, no. 3, pp. 171–179, 2020.
- [42] S. Sarkar, “Sentiment Analysis for Mental Health,” Kaggle Dataset, 2021.
Available: <https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health>
- [43] P. Cuijpers, A. Beekman, and J. Reynolds, “Preventing depression: A global priority,” *JAMA*, vol. 307, no. 10, pp. 1033–1034, 2012.
- [44] S. Minaee, et al., “Deep learning–based text classification: A comprehensive review,” *ACM Computing Surveys*, vol. 54, no. 3, 2021.
- [45] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

- [46] T. Mikolov, K. Chen, G. Corrado, and J. Dean, “Efficient estimation of word representations in vector space,” *Proceedings of ICLR*, 2013.
- [47] Z. Huang, W. Xu, and K. Yu, “Bidirectional LSTM-CRF models for sequence tagging,” *arXiv preprint arXiv:1508.01991*, 2015.
- [48] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, “Dropout: A simple way to prevent neural networks from overfitting,” *Journal of Machine Learning Research*, vol. 15, pp. 1929–1958, 2014.
- [49] Google Research, “Google Colaboratory,” 2023. Available: <https://colab.research.google.com>
- [50] R. Kohavi, “A study of cross-validation and bootstrap for accuracy estimation and model selection,” *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 1137–1143, 1995.
- [51] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *Proceedings of the International Conference on Learning Representations (ICLR)*, 2015.
- [52] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.
- [53] T. Saito and M. Rehmsmeier, “The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets,” *PLoS ONE*, vol. 10, no. 3, 2015.
- [54] M. Shatte, D. Hutchinson, and S. Teague, “Machine learning in mental health: A scoping review,” *Psychological Medicine*, vol. 49, no. 9, pp. 1426–1448, 2019.
- [55] N. Tadesse, H. Lin, B. Xu, and L. Yang, “Detection of depression-related posts in Reddit social media forum,” *IEEE Access*, vol. 7, pp. 44883–44893, 2019.
- [56] L. Bendebane, et al., “A multi-class deep learning approach for early detection of depression and anxiety from tweets,” *Algorithms*, vol. 16, no. 12, 2023.
- [57] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, “Dropout: A simple way to prevent neural networks from overfitting,” *Journal of Machine Learning Research*, vol. 15, pp. 1929–1958, 2014.
- [CU1] A. Sharma, R. Kaur, and S. Gupta, “Machine learning-based mental health prediction using survey data,” *International Journal of Innovative Technology and Exploring Engineering*, 2020. (Chandigarh University)
- [CU2] P. Verma and N. Singh, “Analysis of depression detection techniques using supervised learning models,” *Journal of Information and Optimization Sciences*, 2021. (Chandigarh University)

[CU3] S. Mehta, A. Bansal, and K. Sharma, “Deep learning approaches for sentiment and mental health analysis from social media data,” *International Journal of Advanced Computer Science and Applications*, 2022. (Chandigarh University)

[CU4] R. Kumar and M. Jain, “Explainable artificial intelligence for healthcare prediction systems,” *International Journal of Engineering Research & Technology*, 2023. (Chandigarh University)

[CU5] N. Arora, S. Malhotra, and V. Aggarwal, “A hybrid deep learning framework for mental health risk assessment,” *Springer Lecture Notes in Networks and Systems*, 2024. (Chandigarh University)